

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 3: Univariate Time Series (2)

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Estimation of Univariate Time Series Models

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	2	HB139 .E55 2015
Verbeek	A Guide to Modern Econometrics	8.7, 8.8	HB139 .V465 2012

Univariate Autoregressive Moving Average Models

ARMA(1, 0)

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$$

ARMA(0, 1)

$$y_t = a_0 + \varepsilon + \beta_1 \varepsilon_{t-1}$$

ARMA(1, 1)

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1}$$

ARMA(2, 2)

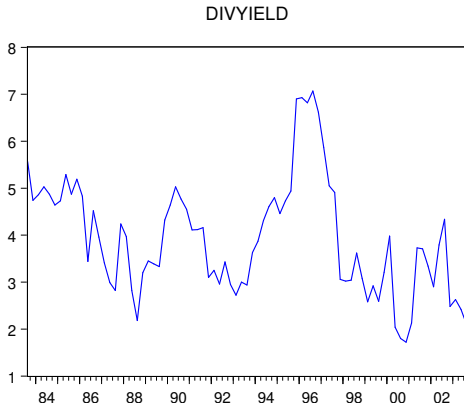
$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \beta_2 \varepsilon_{t-2}$$

ACF and PACF

The ACF and PACF can help us identify the model structure.

MODEL	ACF	PACF
ARMA(1, 0)	Decays to zero	One non-zero peak
ARMA(0, 1)	One non-zero peak	Decays to zero
ARMA(1, 1)	Decays to zero	Decays to zero
ARMA(2, 0)	Decays to zero	Two non-zero peaks

Coles Myers Dividend Yield Series, 1983Q3–2003Q3



Does the plot of the series show any apparent trend?

Model Identification

Date: 08/08/16 Time: 14:07

Sample: 1983Q3 2003Q3

Included observations: 81

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 0.818	0.818	56.284	0.000
		2 0.662	-0.023	93.580	0.000
		3 0.548	0.038	119.44	0.000
		4 0.433	-0.062	135.85	0.000
		5 0.345	0.010	146.35	0.000
		6 0.195	-0.245	149.76	0.000
		7 0.070	-0.043	150.21	0.000
		8 -0.029	-0.068	150.29	0.000
		9 -0.065	0.126	150.68	0.000
		10 -0.095	-0.046	151.54	0.000
		11 -0.204	-0.237	155.55	0.000
		12 -0.280	-0.056	163.21	0.000
		13 -0.324	-0.029	173.57	0.000
		14 -0.360	-0.103	186.59	0.000
		15 -0.323	0.155	197.20	0.000
		16 -0.297	0.032	206.31	0.000
		17 -0.246	0.100	212.64	0.000
		18 -0.210	-0.108	217.33	0.000
		19 -0.155	0.017	219.93	0.000
		20 -0.088	-0.031	220.77	0.000
		21 -0.023	0.124	220.83	0.000
		22 0.040	-0.018	221.02	0.000
		23 0.028	-0.131	221.11	0.000
		24 0.001	-0.114	221.11	0.000

The Q -Statistic

White noise has zero autocorrelation.

Test if a process is white noise using the Ljung-Box (1978) Q -statistic.

- joint test that the first K autocorrelations ($\rho_k = \gamma_k/\gamma_0$) are not significant;
- $H_0 : \rho_1 = \rho_2 = \dots = \rho_K = 0$ versus $H_1 : ?$

Test statistic computed as

$$Q_K = T(T+2) \sum_{k=1}^K \frac{r_k^2}{T+k} \sim \chi_K^2,$$

where r_k is the sample autocorrelation at lag k , and K is the number of autocorrelation lags being tested.

Model Identification for the Coles Myers Dividend Yield Data

The Q -statistic **rejects the null** at all values $K = 1, \dots, 24$ (and beyond).

The SACF is decaying.

The SPACF has a single non-zero peak.

What is the **most suitable** model suggested?

Good practice: fit a range of models around what appears to be “most suitable”.

Estimation and Diagnostics

Recall that a **pure** AR model (i.e. with no MA lags, or equivalently $\text{ARMA}(p, 0)$) can be estimated using OLS.

Including MA lags requires either **Non-linear Least Squares** (NLLS) or **Maximum Likelihood** (MLE).

NLLS chooses parameters $\alpha_0, \dots, \alpha_p, \beta_1, \dots, \beta_q, \sigma_\varepsilon^2$ to **minimize** the **non-linear sum of squares**.

MLE chooses parameters $\alpha_0, \dots, \alpha_p, \beta_1, \dots, \beta_q, \sigma_\varepsilon^2$ to **maximize** the **likelihood**.

- A **likelihood** is the **joint density** of the observed data (e.g. $y_1, \dots, y_T$) as a function of the model parameters.
- Assuming a distribution for the errors implies a **likelihood** for data.
 - See Appendix 2.1 of Enders for the form of the likelihood (and log-likelihood) of the $\text{ARMA}(p, q)$.

Estimation and Diagnostics

AR(1)

Dependent Variable: DIVYIELD
 Method: ARMA Maximum Likelihood (BFGS)
 Date: 03/10/16 Time: 15:42
 Sample: 1983Q3 2003Q3
 Included observations: 81
 Convergence achieved after 5 iterations
 Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.916264	0.445945	8.781954	0.0000
AR(1)	0.850219	0.072178	11.77952	0.0000
SIGMASQ	0.434378	0.051893	8.370618	0.0000
R-squared	0.705297	Mean dependent var	3.927654	
Adjusted R-squared	0.697741	S.D. dependent var	1.221629	
S.E. of regression	0.671628	Akaike info criterion	2.093954	
Sum squared resid	35.18461	Schwarz criterion	2.182637	
Log likelihood	-81.80513	Hannan-Quinn criter.	2.129535	
F-statistic	93.33682	Durbin-Watson stat	1.917667	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.85			

MA(1)

Dependent Variable: DIVYIELD
 Method: ARMA Maximum Likelihood (BFGS)
 Date: 03/10/16 Time: 15:40
 Sample: 1983Q3 2003Q3
 Included observations: 81
 Convergence achieved after 7 iterations
 Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.932779	0.166361	23.63996	0.0000
MA(1)	0.790423	0.077826	10.15624	0.0000
SIGMASQ	0.690675	0.102480	6.739628	0.0000
R-squared	0.531414	Mean dependent var	3.927654	
Adjusted R-squared	0.519398	S.D. dependent var	1.221629	
S.E. of regression	0.846900	Akaike info criterion	2.553966	
Sum squared resid	55.94466	Schwarz criterion	2.642650	
Log likelihood	-100.4356	Hannan-Quinn criter.	2.589547	
F-statistic	44.22903	Durbin-Watson stat	1.324738	
Prob(F-statistic)	0.000000			
Inverted MA Roots	-.79			

Criteria for Model Selection

We can use **information criteria** to choose amongst models in a more formal way.

Two commonly used information criteria are: **Akaike's Information Criterion** (AIC) and **Bayesian Information Criterion** (BIC, or sometimes SC, SBC, SBIC).

All Information criteria quantify a **penalty** for lack of fit and / or over-parameterizations.

- When comparing models using information criteria, **lower is better**.
- The trade off is **fit** versus **parsimony**.

Both AIC and BIC are *likelihood* based:

$$\text{AIC} = \ln \hat{\sigma}_\varepsilon^2 + 2 \frac{p + q + 1}{T} \qquad \text{BIC} = \ln \hat{\sigma}_\varepsilon^2 + \ln T \frac{p + q + 1}{T}.$$

They both sum the log of the **estimated residual variance** (i.e. measure of fit) and estimate of the number of **free parameters**.

Model Identification for the Coles Myers Dividend Yield Data

The AIC is 2.554 for MA(1) and 2.094 for AR(1). Which is better?

The BIC is 2.642 for MA(1) and 2.182 for AR(1). Which is better?

Do the AIC and BIC agree on the preferred specification? How does this compare to what was suggested by the SACF and SPACF?

Analyzing the Residuals

Date: 08/08/16 Time: 15:54

Sample: 1983Q3 2003Q3

Included observations: 81

Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 0.024	0.024	0.0495	
		2 -0.060	-0.061	0.3578	0.550
		3 0.037	0.040	0.4742	0.789
		4 0.038	0.033	0.6029	0.896
		5 0.179	0.183	3.4410	0.487
		6 -0.094	-0.104	4.2337	0.516
		7 -0.061	-0.037	4.5723	0.600
		8 -0.166	-0.203	7.1188	0.417
		9 -0.029	-0.028	7.1990	0.515
		10 0.163	0.131	9.7296	0.373
		11 -0.090	-0.042	10.511	0.397
		12 -0.122	-0.084	11.955	0.367
		13 -0.012	0.023	11.970	0.448
		14 -0.144	-0.209	14.046	0.371
		15 -0.010	-0.061	14.055	0.446
		16 -0.107	-0.113	15.233	0.435
		17 0.020	0.079	15.274	0.505
		18 -0.073	-0.055	15.836	0.535
		19 0.006	0.081	15.841	0.604
		20 0.046	-0.066	16.075	0.652
		21 -0.033	0.004	16.194	0.705
		22 0.224	0.169	21.917	0.404
		23 0.056	0.040	22.286	0.443
		24 0.133	0.175	24.361	0.384

Date: 08/08/16 Time: 15:56

Sample: 1983Q3 2003Q3

Included observations: 81

Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 0.318	0.318	8.5070	
		2 0.582	0.535	37.340	0.000
		3 0.289	0.050	44.524	0.000
		4 0.316	-0.065	53.262	0.000
		5 0.246	0.049	58.612	0.000
		6 0.101	-0.142	59.522	0.000
		7 0.075	-0.130	60.034	0.000
		8 -0.060	-0.105	60.367	0.000
		9 -0.026	0.010	60.432	0.000
		10 -0.063	0.078	60.812	0.000
		11 -0.109	-0.041	61.947	0.000
		12 -0.236	-0.259	67.389	0.000
		13 -0.139	0.032	69.301	0.000
		14 -0.327	-0.146	80.017	0.000
		15 -0.136	0.026	81.910	0.000
		16 -0.263	0.064	89.041	0.000
		17 -0.105	0.121	90.197	0.000
		18 -0.168	0.007	93.214	0.000
		19 -0.105	-0.062	94.411	0.000
		20 -0.013	0.035	94.431	0.000
		21 -0.080	-0.054	95.145	0.000
		22 0.126	0.145	96.949	0.000
		23 -0.054	-0.077	97.285	0.000
		24 0.077	-0.085	97.980	0.000

Note: when analyzing **residuals** of an $ARMA(p, q)$ the Q -statistic for K autocorrelations has distribution χ^2_{K-p-q} .

Inference With Time-Series Data

What can we do with an estimated $\text{ARMA}(p, q)$ for a sample $y_1, \dots, y_T$?

Forecast: predict future values $y_{T+1}, \dots, y_{T+h}$.

Impulse Response Functions: analyze response of variables to unforeseeable shocks.

More possibilities for multivariate dynamic systems.

Lag Operator Notation

To discuss forecasting and impulse response function, it helps to define some notation.

Definition.

The **lag operator** L applied to a stochastic process $\{y_t\}$ transforms a realisation at time t into a realisation at time $t - 1$, i.e.

$$y_{t-1} = Ly_t.$$

This helps us write polynomials in the lag operator:

- $a(L) = 1 - a_1L - \cdots - a_pL^p,$
- $\beta(L) = 1 + \beta_1L + \cdots + \beta_qL^q.$

Then, the ARMA(p, q) can be concisely expressed as $a(L)y_t = \beta(L)\varepsilon_t$.

AR and MA representations of the ARMA(p, q)

If the ARMA(p, q) is **stationary**, then $a(L)$ can be *inverted* to produce the pure MA representation $y_t = \theta(L)\varepsilon_t$, where

$$\theta(L) = \frac{\beta(L)}{a(L)}.$$

We will also refer to this as the **Wold** representation.

If the ARMA(p, q) is **invertible**, then $\beta(L)$ can be *inverted* to produce the pure AR representation $\phi(L)y_t = \varepsilon_t$, where

$$\phi(L) = \frac{a(L)}{\beta(L)}.$$

Forecasting

Recall that given observations $y_1, \dots, y_T$, the **optimal predictor** of a future value y_{T+h} is the conditional expectation $\hat{y}_{T+h} \equiv E(y_{T+h} \mid y_1, \dots, y_T)$.

Given parameter estimates of a $\text{ARMA}(p, q)$, point forecasts are generated recursively using estimated residuals $\{\hat{\varepsilon}_t\}$:

$$\hat{y}_{T+1} = a_0 + a_1 y_T + \dots + a_p y_{T-p+1} + \beta_1 \hat{\varepsilon}_T + \dots + \beta_q \hat{\varepsilon}_{T-q+1},$$

$$\hat{y}_{T+2} = a_0 + a_1 \hat{y}_{T+1} + \dots + a_p y_{T-p+2} + \beta_2 \hat{\varepsilon}_T + \dots + \beta_q \hat{\varepsilon}_{T-q+2},$$

$$\vdots$$

$$\hat{y}_{T+q+1} = a_0 + a_1 \hat{y}_{T+q} + \dots + a_p y_{T-p+q+1}, \quad \text{if } p > q$$

$$\vdots$$

$$\hat{y}_{T+h} = a_0 + a_1 \hat{y}_{T+h-1} + \dots + a_p \hat{y}_{T-p+h}, \quad h > \max\{p, q\}.$$

Forecast Uncertainty

But point forecasts on their own are difficult to interpret.

To make forecasts useful, we need to quantify **uncertainty**. Two sources of uncertainty:

- Uncertainty due to **unknown future**, i.e. future errors $\varepsilon_{T+1}, \dots, \varepsilon_{T+h}$.
- Uncertainty due to **parameter estimation**, i.e. using $\hat{a}_0, \dots, \hat{a}_p, \hat{\beta}_1, \dots, \hat{\beta}_q$ instead of the “true” $a_0, \dots, a_p, \beta_1, \dots, \beta_q$.

If all parameters were known, the **forecast error variance** would be:

$$\text{Var}(y_{T+h} - \hat{y}_{T+h}) = \sigma_\varepsilon^2 (1 + \theta_1^2 + \dots + \theta_{h-1}^2).$$

Example: For an AR(1), $\text{Var}(y_{T+h} - \hat{y}_{T+h}) = \sigma_\varepsilon^2 \frac{1-a_1^{2h}}{1-a_1^2}$.

Question: what happens as $h \rightarrow \infty$?

Impulse Response Functions

Policy is often concerned with **how a variable responds to a shock**.

- If we have a positive, unanticipated shock to GDP, how will GDP respond in the future?
- We consider the **impulse response** path: the difference between the expected path with the shock and the expected path without the shock.

The **impulse response function** (IRF) is the partial derivative $\frac{\partial y_{t+h}}{\partial \varepsilon_t}$ for all $h \geq 0$. In a linear model, this is equivalent to

$$E(y_{t+h} \mid \varepsilon_t = 1, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots) - E(y_{t+h} \mid \varepsilon_t = 0, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots) = \theta_h.$$

Estimating Impulse Response Functions

Given an estimated ARMA(p, q), the IRF can be estimated by either

- iteratively computing $\hat{y}_{t+h}$ for $\varepsilon_t = 1$ and iteratively computing $\bar{y}_{t+h}$ for $\varepsilon_t = 0$, then setting $\text{IR}(h) = \hat{y}_{t+h} - \bar{y}_{t+h}$, or
- by computing $\theta(L) = \beta(L)/a(L)$.

Note that for IRFs **deterministic terms** (such as a_0) **do not matter**—we can ignore them.

$\theta(L)$ can be derived from $a(L)$ and $\beta(L)$ using the **method of undetermined coefficients** (see Enders pp. 33–42).

Uncertainty in IRFs is *only* due to **uncertainty in parameter estimation**.

Example: ARMA(2, 1)

Suppose we estimate the model

$$y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t + \beta_1 \varepsilon_{t-1},$$

$$y_t = 1.2y_{t-1} - 0.7y_{t-2} + \varepsilon_t + 0.3\varepsilon_{t-1}.$$

In this case, $a(L) = 1 - 1.2L + 0.7L^2$ and $\beta(L) = 1 + 0.3L$.

We wish to find

$$\theta(L) = (1 + 0.3L)/(1 - 1.2L + 0.7L^2) = 1 + \theta_1 L + \theta_2 L^2 + \dots$$

Method of Undetermined Coefficients for the ARMA(2, 1)

$$\begin{aligned}(1 + 0.3L)/(1 - 1.2L + 0.7L^2) &= 1 + \theta_1 L + \theta_2 L^2 + \dots, \\ 1 + 0.3L &= (1 - 1.2L + 0.7L^2)(1 + \theta_1 L + \theta_2 L^2 + \dots), \\ 1 + 0.3L &= 1 + \theta_1 L + \theta_2 L^2 + \dots \\ &\quad - 1.2L - 1.2\theta_1 L^2 - 1.2\theta_2 L^3 - \dots \\ &\quad + 0.7L^2 + 0.7\theta_1 L^3 + 0.7\theta_2 L^4 - \dots,\end{aligned}$$

which leads to

$$\begin{aligned}L^0 : \quad & 1 = 1, \\ L^1 : \quad & 0.3 = \theta_1 - 1.2 \quad \Rightarrow \quad \theta_1 = 1.5, \\ L^2 : \quad & 0 = \theta_2 - 1.2\theta_1 + 0.7 \quad \Rightarrow \quad \theta_2 = 1.1, \\ L^3 : \quad & 0 = \theta_3 - 1.2\theta_2 + 0.7\theta_1 \quad \Rightarrow \quad \theta_3 = 0.27, \\ & \dots \\ L^k : \quad & 0 = \theta_k - 1.2\theta_{k-1} + 0.7\theta_{k-2} \quad \Rightarrow \quad \theta_k = 1.2\theta_{k-1} - 0.7\theta_{k-2}.\end{aligned}$$

IRF to a shock $\varepsilon_t = 1$

